

GameFi Economy

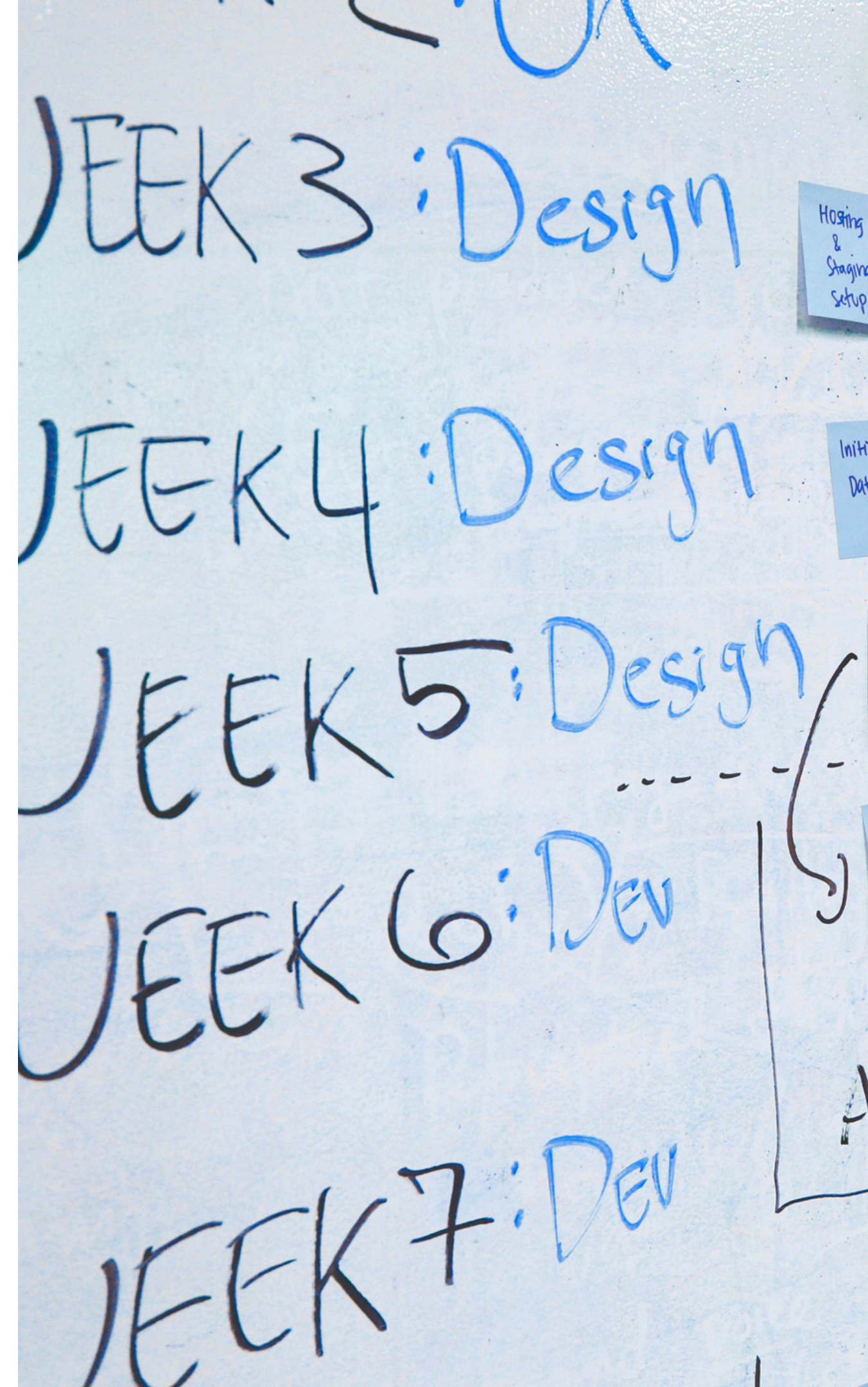
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Blockchain technologies 2

Project Overview

▶ GameFi Economy is a full-stack decentralized protocol developed for the Blockchain Technologies 2 final project.

▶ The protocol implements an on-chain game economy with ERC-1155 game items, Chainlink VRF loot boxes, crafting mechanics, an item rental vault, an AMM, an ERC-4626 vault, DAO governance, frontend dApp, and subgraph integration.





Protocol Architecture

The system consists of several connected components:

- ERC-1155 Gameltems contract for in-game assets
- LootBoxVRF contract for randomized loot drops
- RentalVault for item renting
- Resource AMM for token swaps
- ERC-4626 Vault for yield-style deposits
- Governor and Timelock for DAO governance
- Frontend dApp for user interaction
- Subgraph for indexed protocol data

GameFi Features

The protocol provides core GameFi functionality:

1. Players can hold ERC-1155 items
2. Items can be minted as loot rewards
3. Crafting allows users to burn ingredients and receive new items
4. Rental vault supports depositing and renting items
5. Inventory is displayed directly in the frontend

Example item types include Wooden Sword, Health Potion, Iron Armor, Dragon Scale, and Mystery Chest.



Chainlink VRF Loot Box

The loot box system uses Chainlink VRF to generate verifiable randomness.

1

User opens
a loot box
from the
frontend

2

LootBoxVRF sends
a randomness
request to
Chainlink VRF

3

The contract
selects a
random item
ID

4

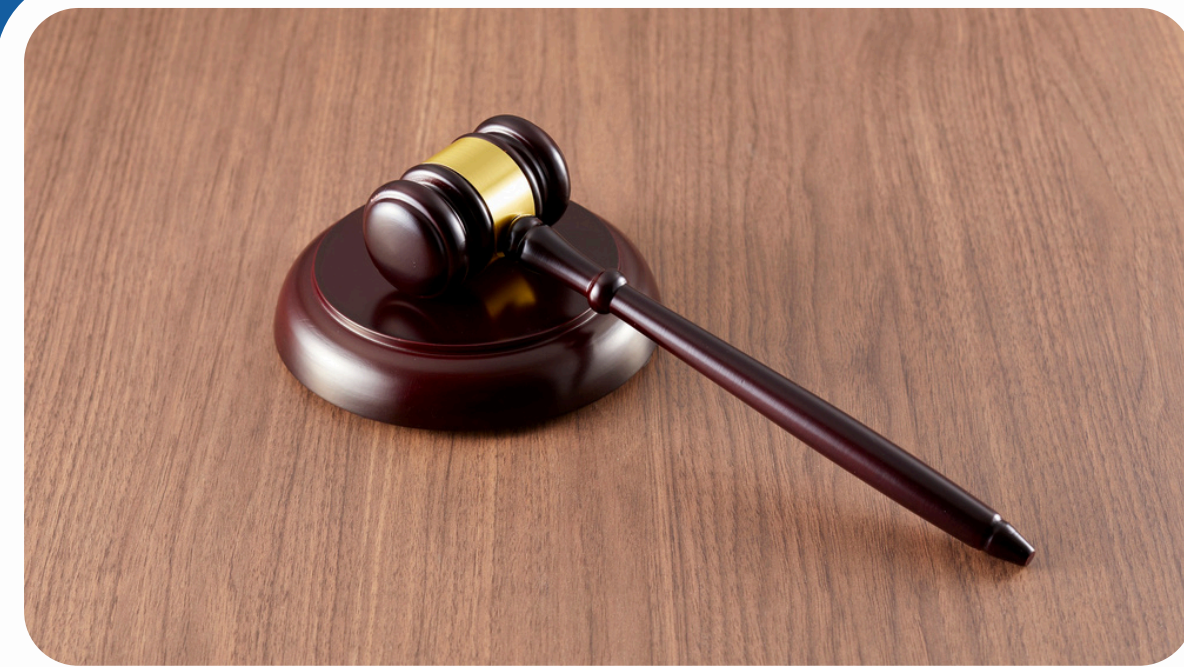
The ERC-
1155 item is
minted to
the user

DeFi and Governance features



DeFi features:

- Constant-product AMM for swaps
- Liquidity reserves
- ERC-4626 vault deposits



Governance features:

- ERC20Votes governance token
- Vote delegation
- Governor contract
- Proposal lifecycle tracking
- Timelock-controlled protocol actions

Frontend dApp

The frontend connects users to the protocol through a browser interface.

The dApp reads on-chain state and sends user transactions through MetaMask.

Main frontend sections:

- Wallet connection and network detection
- Game Hub with inventory, crafting, and loot box
- Rental Vault interactions
- AMM swap interface
- ERC-4626 vault deposit interface
- DAO governance panel
- Subgraph data section



Testing, Deployment, and Result

01

The smart contracts were developed with Foundry and include tests for core protocol logic, revert cases, user actions, and contract interactions. Deployment scripts were used to make the deployment process reproducible instead of deploying contracts manually.

02

The protocol was deployed on Arbitrum Sepolia, an Ethereum Layer 2 testnet. The frontend was connected to the deployed contract addresses, allowing users to interact with the protocol directly from the browser.

03

The final result is a working GameFi Economy dApp that combines ERC-1155 game items, Chainlink VRF loot boxes, crafting, rental mechanics, DeFi components, DAO governance, and indexed data through a subgraph.

THANK YOU!